

Q1 (20 July 2021 Shift 1)

The radiation corresponding to $3 \rightarrow 2$ transition of a hydrogen atom falls on a gold surface to generate photoelectrons. These electrons are passed through a magnetic field of 5×10^{-4} T. Assume that the radius of the largest circular path followed by these electrons is 7 mm, the work function of the metal is:

(Mass of electron = 9.1×10^{-31} kg)

(1) 1.36eV

(2) 1.88eV

(3) 0.16eV

(4) 0.82eV

Q2 (27 July 2021 Shift 1)

In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius $0.5\text{\AA}$ If the speed of electron is 2.2×10^6 m/s, then the current associated with the electron will be $\frac{22}{7} \times 10^{-2}$ mA. [Take π as $\frac{22}{7}$]

Q3 (27 July 2021 Shift 2)

The K_{α} X-ray of molybdenum has wavelength

0.071 nm. If the energy of a molybdenum atoms

with a K electron knocked out is 27.5keV, the energy of this atom when an L electron is knocked out will be keV. (Round off to the nearest

integer)

[$h = 4.14 \times 10^{-15}$ eVs, $c = 3 \times 10^8$ ms $^{-1}$]

Q1 (4)

Q2 (112)

Q3 (10)

Q1

$$\begin{array}{l} 1.51 \text{ ————— } n = 3 \\ 3.4 \text{ ————— } n = \\ 13.6 \text{ ————— } n = 1 \end{array}$$

$$3 \rightarrow 2 \Rightarrow 1.89 \text{ eV}$$

$$5 \times 10^{-4} \text{ T} \quad r = 7 \text{ mm}$$

$$r = \frac{mv}{qB} \Rightarrow mv = qrB$$

$$\begin{aligned} \Rightarrow E &= \frac{p^2}{2m} = \frac{(qRB)^2}{2m} \\ &= \frac{(1.6 \times 10^{-19} \times 7 \times 10^{-3} \times 5 \times 10^{-4})^2}{2 \times 9.1 \times 10^{-31} \text{ Joule}} \\ &= \frac{3136 \times 10^{-52}}{18.2 \times 10^{-31} \times 1.6 \times 10^{-19}} \text{ eV} \end{aligned}$$

$$= 1.077 \text{ eV}$$

$$\text{We know work function} = \text{energy incident} - (\text{KE})_{\text{electron}} \quad \phi = 1.89 - 1.077 = 0.813 \text{ eV}$$

Q2

$$I = \frac{e}{T} = \frac{e\omega}{2\pi} = \frac{eV}{2\pi r}$$

$$I = \frac{1.6 \times 10^{-19} \times 2.2 \times 10^6 \times 7}{2 \times 22 \times 0.5 \times 10^{-10}}$$

$$= 1.12 \text{ mA}$$

$$112 \times 10^{-2} \text{ mA}$$

Q3

$$E_{k_a} = E_k - E_L$$

$$\frac{hc}{\lambda_{k_a}} = E_k - E_L$$

$$\begin{aligned} E_L &= E_k - \frac{hc}{\lambda_{k_a}} \\ &= 27.5 \text{ KeV} - \frac{12.42 \times 10^{-7} \text{ eVm}}{0.071 \times 10^{-9} \text{ m}} \end{aligned}$$

$$E_L = (27.5 - 17.5) \text{ keV}$$

$$= 10 \text{ keV}$$